

"PAPER_{0:D3}" -f [int]# PAPER #120 — UQFF Astronomical Systems Catalog: 24-System Parameter Reference

Author: Daniel T. Murphy

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Title: Complete Catalog of Astronomical Systems Used in UQFF Calculations — Parameters, Verification Sources, and Equation Assignments

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.61$) **Date:** March 2026 **Source:** Grok thread 2fe4fa3e — DeepSearch extraction of "UQFF Equations Across Astrophysical Systems" (393 pages) **Index Slot:** §1.16 UQFF Equation Systems Reference

$$F_U(r, t) = \sum_{i=1}^4 U_{\{gi\}} + U_m + U_A - U_{\{b_i\}}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [\text{SSq}] = 0.57$$

$$U_{\{b_i\}}(r) = \kappa \cdot [\text{SSq}] \cdot \frac{GM}{r^2}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [\text{SSq}] = 0.57, \quad \beta_i = 0.61$$

Abstract

The UQFF framework has been validated and applied across 24 unique astrophysical systems spanning stellar, galactic, nuclear, particle, and cosmological domains. This paper provides a complete parameter catalog for all 24 systems, organized by type, with UQFF equation assignments, calibrated parameter values, verification data sources, and cross-references to existing whitepapers. The catalog is extracted from the 393-page "UQFF Equations Across Astrophysical Systems" corpus (verified Sept 22, 2025) as the authoritative single-document reference for system parameters used in any UQFF calculation.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Stellar and Solar Systems

1.1 Sun (Solar System)

Role in UQFF: Heliosphere modeling via Ug2 bubble, solar wind calibration.

Parameter	Value	Source
M_S	$1.989 \times 10^{30} \text{ kg}$	IAU 2015
R_b (heliosphere bubble radius)	$1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$	Voyager 1 crossing
ω_S (rotation rate)	$2.5 \times 10^{-6} \text{ rad/s}$	Solar sidereal rotation
B_S (surface magnetic field)	$0.0001\text{--}0.4 \text{ T}$	GOES 2025

Parameter	Value	Source
Δ_{sw}	$0.01 = [SSq]/57$	EP-07, PSP CDAWeb E01–E17
v_{sw}	$5 \times 10^5 \text{ m/s}$	EP-07 PSP baseline
ρ_{sw}	$\approx 8 \times 10^{-21} \text{ kg/m}^3$	Parker Solar Probe
H_{SCm}	≈ 1	Heliosphere SCm factor
T_S (surface temperature)	5778 K	IAU

****UQFF equations:**** U_{g2} (bubble), U_m (disk strings), $U_{b,i}$ (expansion) ****Verification:**** EP-07 (PAPER_114), Parker Solar Probe CDAWeb 2025 **** Q_{wave} contribution:**** $Q_{wave,\odot}$ included in 47-system mean $3.97 \times 10^4 \text{ J/m}^3$

1.2 Westerlund 2 (Star Cluster)

Role in UQFF: Triadic master for turbulent outflow modeling.

Parameter	Value	Source
Distance	$\approx 8 \text{ kpc}$	Gaia DR3
Mass	$\sim 10^4 M_\odot$	JWST 2025
Age	$\sim 2 \text{ Myr}$	Membership surveys
SFR	High (active star formation)	JWST 2024
Q_{wave}	$\approx 3.97 \times 10^4 \text{ J/m}^3$ (mean stack)	47-system array

UQFF equations: $F_{U,\text{Triadic}}$ with $n=13$ (plasma), U_m (turbulence) **Verification:** JWST 2025 images, 47-system Q_{wave} stats ($Jarque\text{-}Bera = 8.78$, $p=0.012$) **Cross-reference:** $Q_{wave,47}$ statistical system

2. Black Hole and Galactic Center Systems

2.1 Sagittarius A (Sgr A)

Role in UQFF: SMBH anchor for U_{g4} (BH-star interaction), galactic distance d_g .

Parameter	Value	Uncertainty	Source
M_{bh}	$8.15 \times 10^{36} \text{ kg} = 4.1 \times 10^6 M_\odot$	$\pm 4.68\%$	Gaia DR4 2025
d_g	$2.44 \times 10^{20} \text{ m} = 25,800 \text{ ly}$	$\pm 4.51\%$	Gaia DR4 (PAPER_110)
d_g (UQFF)	$2.55 \times 10^{20} \text{ m} = 27,000 \text{ ly}$	—	UQFF calibrated
ω_g	$7.3 \times 10^{-16} \text{ rad/s}$	—	Galactic rotation curve
U_{g4}	$1.8937 \times 10^{-23} \text{ J/m}^3$	—	PAPER_110 calculated

UQFF equations: U_{g4} , E_{react} (from BH-scale reactivity) **Verification:** EP-06 (PAPER_110), Gaia DR3/DR4 2025 **Note:** The UQFF uses $d_g = 2.55 \times 10^{20} \text{ m}$ (Galactic longitude model) vs Gaia's

2.44×10^{20} m (4.5% difference within observational uncertainty).

2.2 G359.13142-0.20005

Role in UQFF: High-z object for shear map analysis and triadic analogs.

Parameter	Value	Source
Galactic coordinates	$l=359.13^\circ, b=-0.20^\circ$	NISP survey
$\Delta\tau$	≈ 0.05	JWST NISP 2025

UQFF equations: $F_U, \text{Triadic}$, shear maps, $\Delta\tau$ calibration **Verification:** JWST 2025 data, NISP catalog

3. Quasar and Blazar Systems

3.1 3C 273 (Quasar)

Role in UQFF: Asymmetric jet verification of t_n reversals.

Parameter	Value	Source
Redshift z	0.158	Schmidt 1963
Luminosity L	$\sim 10^{46}$ erg/s = 10^{39} W	MNRAS 2025
Jet length	~ 200 kpc (apparent), 65 kpc proper	VLBI
Brightness asymmetry	$> 100:1$	MNRAS 482, 743 (2019)
N (reversal count)	13 sequential t_n reversals	PAPER_115
Asymmetry ratio R	$130 > 100$ target	EP-09 calculation

UQFF equations: $U_{b,i}$ with $\cos(\pi t_n)$, U_m ; ratio $R = |\cos(\pi t_{n1})|/|\cos(\pi t_{n2})|^{1/N}$ **Verification:** EP-09 (PAPER_115), MNRAS 482/743 2019, arXiv:2412.18250

3.2 RACS J0320-35 (Quasar)

Role in UQFF: Navier-Stokes jet fluid verification of $U_{b,i}$ asymmetry.

Parameter	Value	Source
Jet asymmetry ratio R	≈ 1.5	Chandra 2025
τ_{dissip}	9 Gyr	EP-01 calculation
$\cos(\omega t_n)$ reversal	Sign flip confirmed	EP-01

UQFF equations: $U_{b,i}$, Navier-Stokes fluid in jets **Verification:** EP-01 (PAPER_111), Chandra X-ray Observatory 2025

3.3 Blazars (Fermi LAT 4LAC)

Role in UQFF: E_{react} decay rate κ calibration across 3,743 sources.

Parameter	Value	Source
Sample size	3,743 sources (4LAC-DR3)	Fermi HEASARC
Luminosity range	$10^{39}\text{--}10^{47}$ W	4LAC-DR3
κ (measured)	$0.000497 \pm 5\%$ day ⁻¹	EP-05 PAPER_113
8-bin κ error	All < 5%	EP-05
E_{react} range	$10^{39}\text{--}10^{47}$ W/m ³ (8 L bins)	EP-05

UQFF equations: $E_{\text{react}} = 10^{46} e^{-\kappa t}$, $L \propto E_{\text{react}}$ by blazar class
Verification: EP-05 (PAPER_113), Fermi LAT 4LAC-DR4, HEASARC 2025

4. Galaxy Cluster and Radio Systems

4.1 PLCK G287.0+32.9 (Galaxy Cluster)

Role in UQFF: Double radio relic calibration for triadic systems.

Parameter	Value	Source
Type	Galaxy cluster (double relic)	Planck PSZ2
$M_{500,X}$	Calibrated to triadic mass scale	PSZ2
Q_{wave}	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$, $M_{500,X}$ calibration **Verification:** Planck/PSZ2 catalog

4.2 ASKAP J1832-0911 (Radio Transient)

Role in UQFF: Transient source for Q_{wave} array updates and triadic analogs.

Parameter	Value	Source
Survey	ASKAP 2025	ASKAP
Classification	Radio transient	ASKAP
Q_{wave}	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$ transient variant **Verification:** ASKAP 2025 survey

4.3 PSZ2 G181.06+48.47 (Galaxy Cluster)

Role in UQFF: Low-mass merger for double relic analogs.

Parameter	Value	Source
$M_{500,X}$	$2.57 \times 10^{14} M_{\odot}$	Planck 2025
Merger type	Low-mass merger	Planck
Q_{wave}	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$, merger relic analog **Verification:** Planck 2025, PSZ2 catalog

5. Transient and Merger Systems

5.1 GW170817 (BNS Merger)

Role in UQFF: β_i calibration via r-process ejecta fraction.

Parameter	Value	Source
Event date	August 17, 2017	LIGO/Virgo
Total merger mass	$\approx 2.8 M_\odot$	LIGO 2017
Ejecta mass M_{ej}	$\approx 0.05 M_\odot$ total	Spectral fit
Dynamical fraction	$40\% (\approx 1 - \beta_i)$	EP-11
Ejecta velocity	$0.1c$ (blue), $0.3c$ (red)	Kilonova spectral
Y_e	≈ 0.1 (neutron-rich)	EP-11 calibration
r-process fraction $A > 140$	$\approx 95\%$ solar	EP-11
$U_{b,i}$ (calculated)	$\approx 2.3 \times 10^{-3} M_\odot/\text{Gyr}$	EP-11

UQFF equations: $U_{b,i}$ feeding outflows, $Y_e = 1/(1+e^{[SCm]/[UA]})$ Verification: EP-11 (PAPER_109), LIGO/Virgo 2017 + 2025 NR simulations

5.2 AT2024tvd (Astrophysical Transient)

Role in UQFF: NS merger analog for triadic system update (2024).

Parameter	Value	Source
Discovery	2024	ATel/ZTF
Classification	NS merger analog	arXiv 2506.04440
Q_{wave} update	Extends 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$ transient variant Verification: arXiv:2506.04440, ZTF 2024

6. Exoplanet Systems

6.1 TOI 1227 b (Exoplanet)

Role in UQFF: $U_{b,i}$ calibration for planetary mass loss.

Parameter	Value	Source
Mass loss rate	$\approx 10^{12} \text{ g/s}$	TESS/JWST 2025
System	Young transiting exoplanet	TESS
Q_{wave}	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ (atmospheric escape), $F_{U,\text{Triadic}}$ for mass loss dynamics Verification: TESS/JWST 2025

7. Nebula and Star Formation Systems

7.1 Pillars of Creation (Nebula)

Role in UQFF: Buoyancy refinement for star-forming outflows in Ug2/Ub_i.

Parameter	Value	Source
Location	M16 (Eagle Nebula)	—
Distance	\approx 6,500 ly	Gaia
Type	Active star formation, pillar instability	JWST 2022/2024
Q_wave	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ buoyancy refinements, U_{g2} bubble, alpha-conjugate clustering
Verification: JWST 2025, EP-12 alpha BEC (PAPER_107)

7.2 Nebular Dynamics (Generic)

Role in UQFF: Dust/gas stability modeling via U_i .

Parameter	Value	Source
Context	UQFF Drawing 32 Nebular system	Internal
Stability variable	$U_i = \frac{\lambda_i}{\rho_{vac, [SCm]}} \frac{\rho_{vac, [UA]}}{\omega_s} \cos(\pi t_n)$	Framework

UQFF equations: U_i (universal inertia for dust/gas stability)

8. Nuclear and Particle Systems

8.1 Pb-206 (Nuclear System)

Role in UQFF: Nuclear binding energy ladder, [SSq] calibration.

Parameter	Value	Source
n (level)	$8.2 (\Delta n = 0.21)$	EP-04
S_n (neutron separation)	$\approx 2 \times [SSq] \times E_8$ (3.5% error)	EP-04
Level spacing	$\sim 10 \text{ MeV} = 10^{-12} \text{ J}$	ENSDF/NNDC 2025
Polynomial fit degree	$n = 8 (R^2 \approx 0.95)$	EP-04
Shell levels	10+ levels in NNDC database	ENSDF

UQFF equations: $E_n = E_0 \times 10^n$ (n=8), [SSq] = 0.57 calibration point
Verification: EP-04 (PAPER_117), ENSDF/NNDC 2025

8.2 ¹²C Hoyle State (Nuclear BEC)

Role in UQFF: Bose term N_B calibration in U_m .

Parameter	Value	Source
Energy	7.65 MeV	Hoyle 1954
Configuration	3-alpha BEC ($N_B = 3$)	Tohsaki AMD
Condensate fraction	70–90%	Tohsaki et al. 2001
T_C (nuclear BEC)	$\approx 1.2 \times 10^6$ K	EP-12 calculation
LENR ΔT_C shift	≈ 300 K	EP-12 UQFF
^{16}O analog	4-alpha BEC ($N_B = 4$)	Funaki et al.

UQFF equations: N_B term in U_m , LENR T_C shift via E_{react} scaling **Verification:** EP-12 (PAPER_107), Tohsaki et al. arXiv:1103.3940

9. Compact Object Systems

9.1 Magnetar SGR 1745-2900

Role in UQFF: Full g_{Magnetar} equation anchor; B/B_{crit} calibration.

Parameter	Value	Source
Type	Soft gamma repeater magnetar	Chandra/Swift
B field	$\sim 10^{10}\text{--}10^{11}$ T	Swift 2025
B_{crit}	4.4×10^{13} T	QED critical
B/B_{crit}	$\sim 10^{-3}$ to 10^{-2}	Calibrated
Mass M_{ns}	$\approx 1.4 M_{\odot}$	Standard NS
Galactic center distance	~ 0.1 pc from Sgr A*	VLBI

UQFF equations: $g_{\text{Magnetar}}(r,t) = (GM/r^2)(1+H(z)\cdot t)(1-B/B_{\text{crit}}) + (GM_{\text{BH}}/r_{\text{BH}}^2) + U_{g1}+U_{g2}+U_{g3}+U_{g4} + \Lambda c^2/3$ **Verification:** Chandra/Swift 2025, PAPER013 (*spin-down*), PAPER121 (full equation) **See:** PAPER_121 for the complete 50+ term expansion

10. Galactic Structural Systems

10.1 Galactic Disk

Role in UQFF: Stability analysis for $U_{b,i}$, galactic rotation curve.

Parameter	Value	Source
ω_g	7.3×10^{-16} rad/s	Galactic rotation
v_{gal}	220 km/s	Milky Way rotation curve
r_{8kpc}	8 kpc = 2.47×10^{20} m	IAU 2012

UQFF equations: $U_{b,i}$ stability, U_{g4} BH interaction **Verification:** Gaia DR4 2025

11. AGN and Cosmic Ray Systems

11.1 LLAGNs (Low-Luminosity AGNs)

Role in UQFF: CRP neutrino flux at \sim IceCube background level.

Parameter	Value	Source
Neutrino flux	\sim IceCube background 10^{-18} GeV ¹ cm ² s ⁻¹ sr ⁻¹	IceCube 2022
Dominant process	pp collisions < 0.1 PeV	EP-10
SED peak	0.05 PeV	EP-10 PAPER_108
Spectral index	\Gamma \approx 2.37	IceCube HESE

UQFF equations: CRP term $\sum D_E \frac{\partial^2 n}{\partial p^2} \cdot e^{-\gamma t}$, pp/p? SED Verification: EP-10 (PAPER_108), IceCube 2022

11.2 High-Energy Cosmic Ray Sources

Role in UQFF: CRP p_{max} calibration.

Parameter	Value	Source
p_{max}	10^{16} eV	Fermi/Chandra 2025
n(p) \propto	$p^{-2.2} e^{-p/p_{max}}$	Fokker-Planck solution
D_E \propto	$E^{0.5}$ (Kolmogorov turbulence)	CRP physics

UQFF equations: Fokker-Planck equation, CRP term in F_U Verification: Fermi/Chandra/Parker 2025

12. Cosmological Systems

12.1 Solar Flares

Role in UQFF: U_{g1} dipole calibration, B_s = 0.4 T upper bound.

Parameter	Value	Source
B_s (flare)	0.4 T	GOES 2025
B_s (quiet)	0.0001 T	GOES baseline

UQFF equations: U_{g1} dipole term Verification: GOES 2025

12.2 Intergalactic Medium

Role in UQFF: \rho_{vac} ratio calibration for vacuum hierarchy.

Parameter	Value	Source
\rho_{vac} ratios	\sim 10^{-38} (low/high components)	EP-08
\lambda_{vac} range	\sim 10^{-9} J/m^3 (DE level)	JCAP 2025
DM density	\sim 0.47 GeV/cm^3 = 8.4 \times 10^{-25} J/m^3	JCAP01(2025)021

Parameter	Value	Source
Primordial DM	$\sim 10^{-26} \text{ J/m}^3$	JCAP07(2025)033

UQFF equations: $\lambda_{\text{vac}} = \sum(f_i E_i)/V, \rho_{\text{vac}}$ alignment **Verification:** EP-08 (PAPER_118), JCAP 2024/2025

12.3 Quasar Jets (General Class)

Role in UQFF: Fluid/Navier-Stokes jet modeling with $U_{b,i}$ asymmetry.

Parameter	Value	Source
Jet velocity	$\sim 0.7c-0.99c$	VLBI proper motion
Asymmetry mechanism	$\cos(\omega t_n)$ reversal	UQFF $U_{b,i}$

UQFF equations: $U_{b,i}$, Navier-Stokes in jet fluid **Verification:** Chandra/MNRAS general

13. Qwave47 Statistical Summary

The 47-system Q_{wave} distribution (from the UQFF verification corpus) encompasses a subset of the 24 systems above plus additional intermediate systems:

Statistic	Value
Mean $\bar{Q}_{\text{wave}}$	$3.97 \times 10^4 \text{ J/m}^3$
Standard deviation	$5.11 \times 10^4 \text{ J/m}^3$
Jarque-Bera	8.78 ($p = 0.012$)
Leptokurtosis	0.037
Distribution	Non-normal (heavy-tailed)
N systems	47 (superset of 24 catalog systems)

The non-normal distribution confirms that UQFF Q_{wave} energy is not uniformly distributed across system types — triadic/turbulent systems (Westerlund 2, Pillars) contribute heavy tails, while individual systems (LLAGNs, nuclear) contribute near-zero floor values.

14. Cross-Reference Table: Systems by Whitepaper

System	Type	Primary Paper	EP
Sun	Stellar	PAPER_114	EP-07
Westerlund 2	Star cluster	Qwave47 corpus	—
Sgr A*	SMBH	PAPER_110	EP-06
G359.13142-0.20005	High-z	JWST corpus	—
3C 273	Quasar	PAPER_115	EP-09
RACS J0320-35	Quasar	PAPER_111	EP-01
Blazars (4LAC)	Blazars	PAPER_113	EP-05
PLCK G287.0+32.9	Galaxy cluster	Triadic corpus	—
ASKAP J1832-0911	Radio transient	Triadic corpus	—

System	Type	Primary Paper	EP
PSZ2 G181.06+48.47	Galaxy cluster	Triadic corpus	—
GW170817	BNS merger	PAPER_109	EP-11
AT2024tvd	Transient	Triadic corpus	—
TOI 1227 b	Exoplanet	Triadic corpus	—
Pillars of Creation	Nebula	Triadic corpus	—
Nebular Dynamics	Generic nebula	Framework	—
Pb-206	Nuclear	PAPER_117	EP-04
¹² C Hoyle State	Nuclear BEC	PAPER_107	EP-12
SGR 1745-2900	Magnetar	PAPER_013/121	—
Galactic Disk	Galactic	PAPER_110	—
LLAGNs	AGN class	PAPER_108	EP-10
CR Sources	High-energy	PAPER_088	—
Solar Flares	Sun	PAPER_114	EP-07
Intergalactic Medium	Cosmological	PAPER_118	EP-08
Quasar Jets	AGN class	PAPER_111/115	EP-01/09

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d_g (UQFF)	2.55×10^{20} m = 27,000 ly	—	UQFF calibrated
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$U_{\{g4\}}$	1.8937×10^{-23} J/m ³	—	PAPER_110 calculated

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$\Delta\tau$	≈ 0.05	JWST NISP 2025

UQFF equations: $F_{\{U, \text{Triadic}\}}$, shear maps, $\Delta\tau$ calibration **Verification:** JWST 2025 data, NISP catalog

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τ_{dissip}	9 Gyr	EP-01 calculation
$\cos(\omega t_n)$ reversal	Sign flip confirmed	EP-01

UQFF equations: $U_{b,i}$, Navier-Stokes fluid in jets **Verification:** EP-01 (PAPER_111), Chandra X-ray Observatory 2025

3.3 Blazars (Fermi LAT 4LAC)

Role in UQFF: E_{react} decay rate κ calibration across 3,743 sources.

Parameter	Value	Source
Sample size	3,743 sources (4LAC-DR3)	Fermi HEASARC
Luminosity range	$10^{39}\text{--}10^{47}$ W	4LAC-DR3
κ (measured)	$0.000497\text{ pm } 5\% \text{ day}^{-1}$	EP-05 PAPER_113
8-bin κ error	All $< 5\%$	EP-05
E_{react} range	$10^{39}\text{--}10^{47}$ W/m ³ (8 L bins)	EP-05

UQFF equations: $E_{\text{react}} = 10^{46} e^{-\kappa t}$, $L \propto E_{\text{react}}$ by blazar class **Verification:** EP-05 (PAPER_113), Fermi LAT 4LAC-DR4, HEASARC 2025

4. Galaxy Cluster and Radio Systems

4.1 PLCK G287.0+32.9 (Galaxy Cluster)

Role in UQFF: Double radio relic calibration for triadic systems.

Parameter	Value	Source
Type	Galaxy cluster (double relic)	Planck PSZ2
$M_{500,X}$	Calibrated to triadic mass scale	PSZ2
Q_{wave}	In 47-system array	Grok corpus

UQFF equations: $F_U, \text{Triadic}$, $M_{500,X}$ calibration **Verification:** Planck/PSZ2 catalog

4.2 ASKAP J1832-0911 (Radio Transient)

Role in UQFF: Transient source for Q_{wave} array updates and triadic analogs.

Parameter	Value	Source
Survey	ASKAP 2025	ASKAP
Classification	Radio transient	ASKAP
Q_wave	In 47-system array	Grok corpus

UQFF equations: $F_{\{U, \text{Triadic}\}}$ transient variant **Verification:** ASKAP 2025 survey

4.3 PSZ2 G181.06+48.47 (Galaxy Cluster)

Role in UQFF: Low-mass merger for double relic analogs.

Parameter	Value	Source
$M_{\{500, X\}}$	$2.57 \times 10^{14} \text{ M?}$	Planck 2025
Merger type	Low-mass merger	Planck
Q_wave	In 47-system array	Grok corpus

UQFF equations: $F_{\{U, \text{Triadic}\}}$, merger relic analog **Verification:** Planck 2025, PSZ2 catalog

5. Transient and Merger Systems

5.1 GW170817 (BNS Merger)

Role in UQFF: β_i calibration via r-process ejecta fraction.

Parameter	Value	Source
Event date	August 17, 2017	LIGO/Virgo
Total merger mass	$\approx 2.8 \text{ M?}$	LIGO 2017
Ejecta mass $M_{\{ej\}}$	$\approx 0.05 \text{ M? total}$	Spectral fit
Dynamical fraction	$40\% (\approx 1 - \beta_i)$	EP-11
Ejecta velocity	$0.1c$ (blue), $0.3c$ (red)	Kilonova spectral
Y_e	≈ 0.1 (neutron-rich)	EP-11 calibration
r-process fraction $A > 140$	$\approx 95\%$ solar	EP-11
$U_{\{b, i\}}$ (calculated)	$\approx 2.3 \times 10^{-3} \text{ M?/Gyr}$	EP-11

UQFF equations: $U_{\{b, i\}}$ feeding outflows, $Y_e = 1/(1+e^{\{[SCm]/[UA]\}})$ **Verification:** EP-11 (PAPER_109), LIGO/Virgo 2017 + 2025 NR simulations

5.2 AT2024tvd (Astrophysical Transient)

Role in UQFF: NS merger analog for triadic system update (2024).

Parameter	Value	Source
Discovery	2024	ATel/ZTF
Classification	NS merger analog	arXiv 2506.04440
Q_wave update	Extends 47-system array	Grok corpus

UQFF equations: $F_{U, \text{Triadic}}$ transient variant **Verification:** arXiv:2506.04440, ZTF 2024

6. Exoplanet Systems

6.1 TOI 1227 b (Exoplanet)

Role in UQFF: $U_{b,i}$ calibration for planetary mass loss.

Parameter	Value	Source
Mass loss rate	$\approx 10^{12}$ g/s	TESS/JWST 2025
System	Young transiting exoplanet	TESS
Q_{wave}	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ (atmospheric escape), $F_{U, \text{Triadic}}$ for mass loss dynamics
Verification: TESS/JWST 2025

7. Nebula and Star Formation Systems

7.1 Pillars of Creation (Nebula)

Role in UQFF: Buoyancy refinement for star-forming outflows in $U_{g2}/U_{b,i}$.

Parameter	Value	Source
Location	M16 (Eagle Nebula)	—
Distance	$\approx 6,500$ ly	Gaia
Type	Active star formation, pillar instability	JWST 2022/2024
Q_{wave}	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ buoyancy refinements, U_{g2} bubble, α -conjugate clustering
Verification: JWST 2025, EP-12 α BEC (PAPER_107)

7.2 Nebular Dynamics (Generic)

Role in UQFF: Dust/gas stability modeling via U_i .

Parameter	Value	Source
Context	UQFF Drawing 32 Nebular system	Internal
Stability variable	$U_i = \frac{\lambda_i}{\rho_{\text{vac}} [\text{SCm}]} \frac{\rho_{\text{vac}} [\text{UA}]}{\omega_s} \cos(\pi t_n)$	Framework

UQFF equations: U_i (universal inertia for dust/gas stability)

8. Nuclear and Particle Systems

8.1 Pb-206 (Nuclear System)

Role in UQFF: Nuclear binding energy ladder, [SSq] calibration.

Parameter	Value	Source
n (level)	8.2 ($\Delta n = 0.21$)	EP-04
S_n (neutron separation)	$\approx 2 \times [\text{SSq}] \times E_8$ (3.5% error)	EP-04
Level spacing	$\sim 10 \text{ MeV} = 10^{-12} \text{ J}$	ENSDF/NNDC 2025
Polynomial fit degree	$n = 8 (R^2 \approx 0.95)$	EP-04
Shell levels	10+ levels in NNDC database	ENSDF

UQFF equations: $E_n = E_0 \times 10^n$ (n=8), [SSq] = 0.57 calibration point Verification: EP-04 (PAPER_117), ENSDF/NNDC 2025

8.2 ¹²C Hoyle State (Nuclear BEC)

Role in UQFF: Bose term N_B calibration in U_m.

Parameter	Value	Source
Energy	7.65 MeV	Hoyle 1954
Configuration	3-alpha BEC (N_B = 3)	Tohsaki AMD
Condensate fraction	70-90%	Tohsaki et al. 2001
T_c (nuclear BEC)	$\approx 1.2 \times 10^6 \text{ K}$	EP-12 calculation
LENR ΔT_c shift	$\approx 300 \text{ K}$	EP-12 UQFF
¹⁶ O analog	4-alpha BEC (N_B = 4)	Funaki et al.

UQFF equations: N_B term in U_m, LENR T_c shift via E_{react} scaling Verification: EP-12 (PAPER_107), Tohsaki et al. arXiv:1103.3940

9. Compact Object Systems

9.1 Magnetar SGR 1745-2900

Role in UQFF: Full g_{Magnetar}(r,t) equation anchor; B/B_{crit} calibration.

Parameter	Value	Source
Type	Soft gamma repeater magnetar	Chandra/Swift
B field	$\sim 10^{10} - 10^{11} \text{ T}$	Swift 2025
B_{crit}	$4.4 \times 10^{13} \text{ T}$	QED critical
B/B_{crit}	$\sim 10^{-3} \text{ to } 10^{-2}$	Calibrated
Mass M_{ns}	$\approx 1.4 \text{ M?}$	Standard NS
Galactic center distance	$\sim 0.1 \text{ pc}$ from Sgr A*	VLBI

UQFF equations: $g_{Magnetar}(r,t) = (GM/r^2)(1+H(z) \cdot t)(1-B/B_{crit}) + (GM_{BH}/r_{BH}^2) + U_{g1}+U_{g2}+U_{g3}+U_{g4} + \Lambda c^2/3$ Verification:

Chandra/Swift 2025, PAPER013 (spin-down), PAPER121 (full equation) **See:** PAPER_121 for the complete 50+ term expansion

10. Galactic Structural Systems

10.1 Galactic Disk

Role in UQFF: Stability analysis for $U_{\{b,i\}}$, galactic rotation curve.

Parameter	Value	Source
ω_g	7.3×10^{-16} rad/s	Galactic rotation
v_{gal}	220 km/s	Milky Way rotation curve
r_{8kpc}	8 kpc = 2.47×10^{20} m	IAU 2012

UQFF equations: $U_{\{b,i\}}$ stability, U_{g4} BH interaction **Verification:** Gaia DR4 2025

11. AGN and Cosmic Ray Systems

11.1 LLAGNs (Low-Luminosity AGNs)

Role in UQFF: CRP neutrino flux at $\sim$ IceCube background level.

Parameter	Value	Source
Neutrino flux	$\sim$ IceCube background 10^{-18} GeV ¹ cm ² s ⁻¹ sr ⁻¹	IceCube 2022
Dominant process	pp collisions < 0.1 PeV	EP-10
SED peak	0.05 PeV	EP-10 PAPER_108
Spectral index	$\gamma \approx 2.37$	IceCube HESE

UQFF equations: CRP term $\sum D_E \partial^2 n / \partial p^2 \cdot e^{-\gamma t}$, pp/p? SED **Verification:** EP-10 (PAPER_108), IceCube 2022

11.2 High-Energy Cosmic Ray Sources

Role in UQFF: CRP $p_{\{max\}}$ calibration.

Parameter	Value	Source
$p_{\{max\}}$	10^{16} eV	Fermi/Chandra 2025
$n(p) \propto$	$p^{-2.2} e^{-p/p_{\{max\}}}$	Fokker-Planck solution
$D_E \propto$	$E^{0.5}$ (Kolmogorov turbulence)	CRP physics

UQFF equations: Fokker-Planck equation, CRP term in F_U **Verification:** Fermi/Chandra/Parker 2025

12. Cosmological Systems

12.1 Solar Flares

Role in UQFF: U_{g1} dipole calibration, $B_s = 0.4$ T upper bound.

Parameter	Value	Source
B_s (flare)	0.4 T	GOES 2025
B_s (quiet)	0.0001 T	GOES baseline

UQFF equations: U_{g1} dipole term **Verification:** GOES 2025

12.2 Intergalactic Medium

Role in UQFF: ρ_{vac} ratio calibration for vacuum hierarchy.

Parameter	Value	Source
ρ_{vac} ratios	$\sim 10^{-38}$ (low/high components)	EP-08
λ_{vac} range	$\sim 10^{-9}$ J/m ³ (DE level)	JCAP 2025
DM density	~ 0.47 GeV/cm ³ = 8.4×10^{-25} J/m ³	JCAP01(2025)021
Primordial DM	$\sim 10^{-26}$ J/m ³	JCAP07(2025)033

UQFF equations: $\lambda_{vac} = \sum(f_i E_i)/V$, ρ_{vac} alignment **Verification:** EP-08 (PAPER_118), JCAP 2024/2025

12.3 Quasar Jets (General Class)

Role in UQFF: Fluid/Navier-Stokes jet modeling with $U_{b,i}$ asymmetry.

Parameter	Value	Source
Jet velocity	$\sim 0.7c-0.99c$	VLBI proper motion
Asymmetry mechanism	$\cos(\omega t_n)$ reversal	UQFF $U_{b,i}$

UQFF equations: $U_{b,i}$, Navier-Stokes in jet fluid **Verification:** Chandra/MNRAS general

13. Qwave47 Statistical Summary

The 47-system Q_{wave} distribution (from the UQFF verification corpus) encompasses a subset of the 24 systems above plus additional intermediate systems:

Statistic	Value
Mean $\bar{Q}_{wave}$	3.97×10^4 J/m ³
Standard deviation	5.11×10^4 J/m ³
Jarque-Bera	8.78 (p = 0.012)
Leptokurtosis	0.037
Distribution	Non-normal (heavy-tailed)
N systems	47 (superset of 24 catalog systems)

The non-normal distribution confirms that UQFF Q_wave energy is not uniformly distributed across system types — triadic/turbulent systems (Westerlund 2, Pillars) contribute heavy tails, while individual systems (LLAGNs, nuclear) contribute near-zero floor values.

14. Cross-Reference Table: Systems by Whitepaper

System	Type	Primary Paper	EP
Sun	Stellar	PAPER_114	EP-07
Westerlund 2	Star cluster	Qwave47 corpus	—
Sgr A*	SMBH	PAPER_110	EP-06
G359.13142-0.20005	High-z	JWST corpus	—
3C 273	Quasar	PAPER_115	EP-09
RACS J0320-35	Quasar	PAPER_111	EP-01
Blazars (4LAC)	Blazars	PAPER_113	EP-05
PLCK G287.0+32.9	Galaxy cluster	Triadic corpus	—
ASKAP J1832-0911	Radio transient	Triadic corpus	—
PSZ2 G181.06+48.47	Galaxy cluster	Triadic corpus	—
GW170817	BNS merger	PAPER_109	EP-11
AT2024tvd	Transient	Triadic corpus	—
TOI 1227 b	Exoplanet	Triadic corpus	—
Pillars of Creation	Nebula	Triadic corpus	—
Nebular Dynamics	Generic nebula	Framework	—
Pb-206	Nuclear	PAPER_117	EP-04
¹² C Hoyle State	Nuclear BEC	PAPER_107	EP-12
SGR 1745-2900	Magnetar	PAPER_013/121	—
Galactic Disk	Galactic	PAPER_110	—
LLAGNs	AGN class	PAPER_108	EP-10
CR Sources	High-energy	PAPER_088	—
Solar Flares	Sun	PAPER_114	EP-07
Intergalactic Medium	Cosmological	PAPER_118	EP-08
Quasar Jets	AGN class	PAPER_111/115	EP-01/09

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